Practical No:7

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Problem Statement: You are given a list of online orders, each with an estimated delivery time in minutes. Write a program to sort these orders using Merge Sort.

Code:

```
def mergeSort(arr):

  if len(arr) > 1:

    mid = len(arr) // 2
    left = arr[:mid]
    right = arr[mid:]

    mergeSort(left)
    mergeSort(right)

    i = j = k = 0

    # Merge the two halves
    while i < len(left) and j < len(right):

      if left[i] < right[j]:
        arr[k] = left[i]
        i += 1
      else:
        arr[k] = right[j]
        j += 1
```

```
            k += 1

        # Copy remaining elements
        while i < len(left):
            arr[k] = left[i]
            i += 1
            k += 1

        while j < len(right):
            arr[k] = right[j]
            j += 1
            k += 1

# Main Program

orders = []

n = int(input("Enter number of online orders: "))

for i in range(n):
    time = int(input("Enter delivery time (minutes): "))
    orders.append(time)

print("\nOriginal Delivery Times:")
print(orders)

mergeSort(orders)
```

```
print("\nSorted Delivery Times (Quickest First):")
print(orders)6
```

Output:

```
Enter number of online orders: 6
Enter delivery time (minutes): 25
Enter delivery time (minutes): 34
Enter delivery time (minutes): 45
Enter delivery time (minutes): 11
Enter delivery time (minutes): 20
Enter delivery time (minutes): 43

Original Delivery Times:
[25, 34, 45, 11, 20, 43]

Sorted Delivery Times (Quickest First):
[11, 20, 25, 34, 43, 45]
```